Vipul Sapra

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EDUCATION

University of Cincinnati, Carl H. Lindner College of Business — **Cincinnati, OH**
Master of Science in Information Systems and Graduate Certificate in Data Science — **April 2022**
Cumulative GPA: 3.95 / 4.0

Pune University, Maharashtra Institute of Technology — **Pune, India**
Bachelor of Engineering in Petrochemical — **June 2017**

TECHNICAL SKILLS

Concepts: Data-Ingestion & ETL Pipelines, Dimensional Data Modeling (Star/Snowflake), Slowly Changing Dimensions (SCD Type 1/2), Workflow Automation & Power Platform Solutions, Predictive Modeling & Machine Learning, Data Quality Audits & Firewalls, Root-Cause Analysis (RCA) & Defect Resolution, Agile Solution Lifecycle & Source Control

Programming Languages & Tools: SQL, Python, PySpark, Azure Data Factory, Azure Synapse, Azure SQL, Databricks, AWS (S3, Redshift), Snowflake, Power BI, Tableau, Power Automate, Power Apps, Airflow, Scikit-Learn, XGBoost, SARIMA, KIRO IDE

Certifications: Data Science Graduate Certificate – University of Cincinnati

PROFESSIONAL EXPERIENCE

Charter Communications — Charlotte, NC

Senior Data, Application & Automation Engineer — **May 2022 – Present**

- **Architected high-performance SQL stored procedures, dimensional** data models (Star/Snowflake schemas), and Slowly Changing Dimensions (SCD Type 1/2) ensuring robust data-ingestion, extraction, transformation, loading, and validation processes supporting enterprise-scale operational workflows.
- **Engineered automated data quality audit frameworks, anomaly** detection systems (95% accuracy), and Power Automate workflow integrations that reduced operational turnaround by over 80% and minimized recurring data discrepancies.
- **Developed advanced predictive models and analytics products** (SARIMA, XGBoost, Scikit-Learn) integrated with Power BI semantic models and executive KOM dashboards, optimizing workforce capacity planning and decision-making by 20%.
- **Deployed SpectrumIQ conversational AI and automated QA testing** frameworks, cutting manual turnaround times by 50% while applying rigorous source-control, documentation, and change-management standards.
- **Spearheaded Agentic AI and prompt engineering initiatives (KIRO IDE)** for automated profiling, accelerating delivery cycles by 40% while maintaining 99.9% data availability SLA through proactive Airflow/One Automation monitoring and RCA postmortems.

Graduate Research & Information Systems Analyst — **May 2022 – Present**

- **Designed relational database schemas and optimized complex** SQL queries to extract, transform, and load large datasets, improving query performance and ensuring data integrity for academic research systems.
- **Built interactive dashboards and reporting solutions using** Tableau and Python, translating technical requirements into actionable insights and maintaining comprehensive technical documentation for stakeholders.
- **Applied structured software development lifecycle (SDLC) methodologies** to test, validate, and deploy modular code components and automated data pipelines in collaborative, multi-disciplinary team environments.

Petrochemical Engineering Analyst — **May 2022 – Present**

- **Analyzed complex operational datasets and process parameters** to identify bottlenecks, troubleshoot system anomalies, and recommend data-driven corrective actions for industrial processes.
- **Managed technical documentation, workflow schedules, and compliance** reports, coordinating closely with cross-functional teams to meet strict project milestones and quality standards.
- **Implemented process automation improvements and data validation checks** to streamline operational workflows, reduce manual errors, and enhance overall system reliability.

Ameritas Insurance — Cincinnati, OH

Data Governance Intern — **September 2021 – April 2022**

- **Co-engineered the enterprise-wide Data Advocacy & Governance Program** across insurance portfolios, establishing foundational frameworks for data democratization and organizational accountability.
- **Architected a comprehensive Data Intelligence ecosystem in Collibra**, establishing a centralized Data Dictionary, automated Data Quality rules, and end-to-end **Data Lineage** to ensure regulatory compliance and transparency for financial and insurance assets.

Accenture — Pune, India

Test Engineering Analyst — **September 2017 – December 2020**

- **Orchestrated end-to-end Quality Assurance and system integration testing** for mission-critical client applications at **Allianz Insurance**, leveraging Agile SDLC methodologies to ensure rapid, zero-defect feature deployments.
- **Spearheaded the Health Insurance module QA strategy**, directing cross-functional efforts for regression testing, test data validation, and the seamless rollout of complex new business functionalities.
- **Drove operational excellence by automating core regression test suites**, significantly increasing engineering productivity and reducing manual testing overhead through scalable automation frameworks.